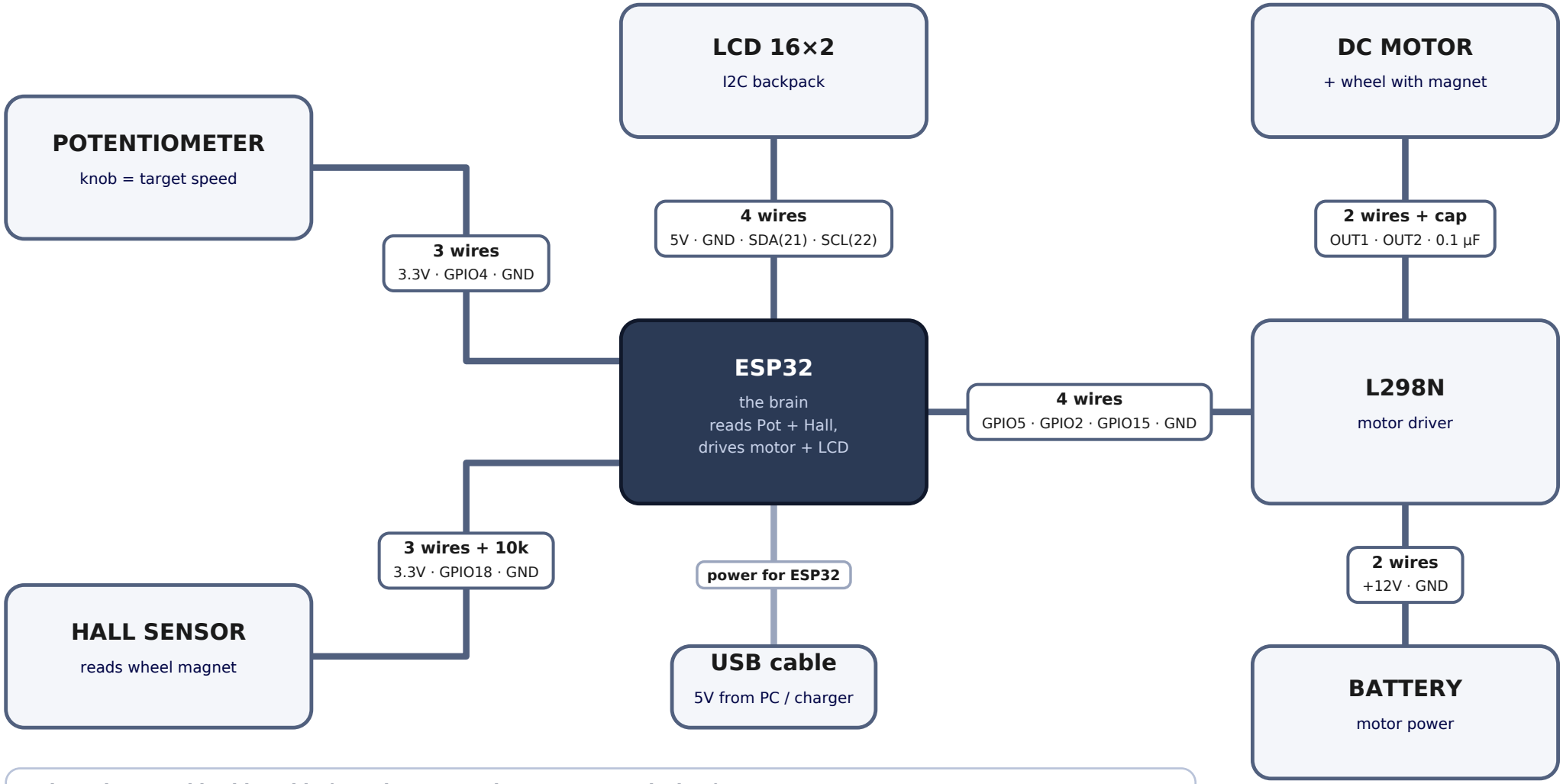
ESP32 RPM Controller - Wiring Guide

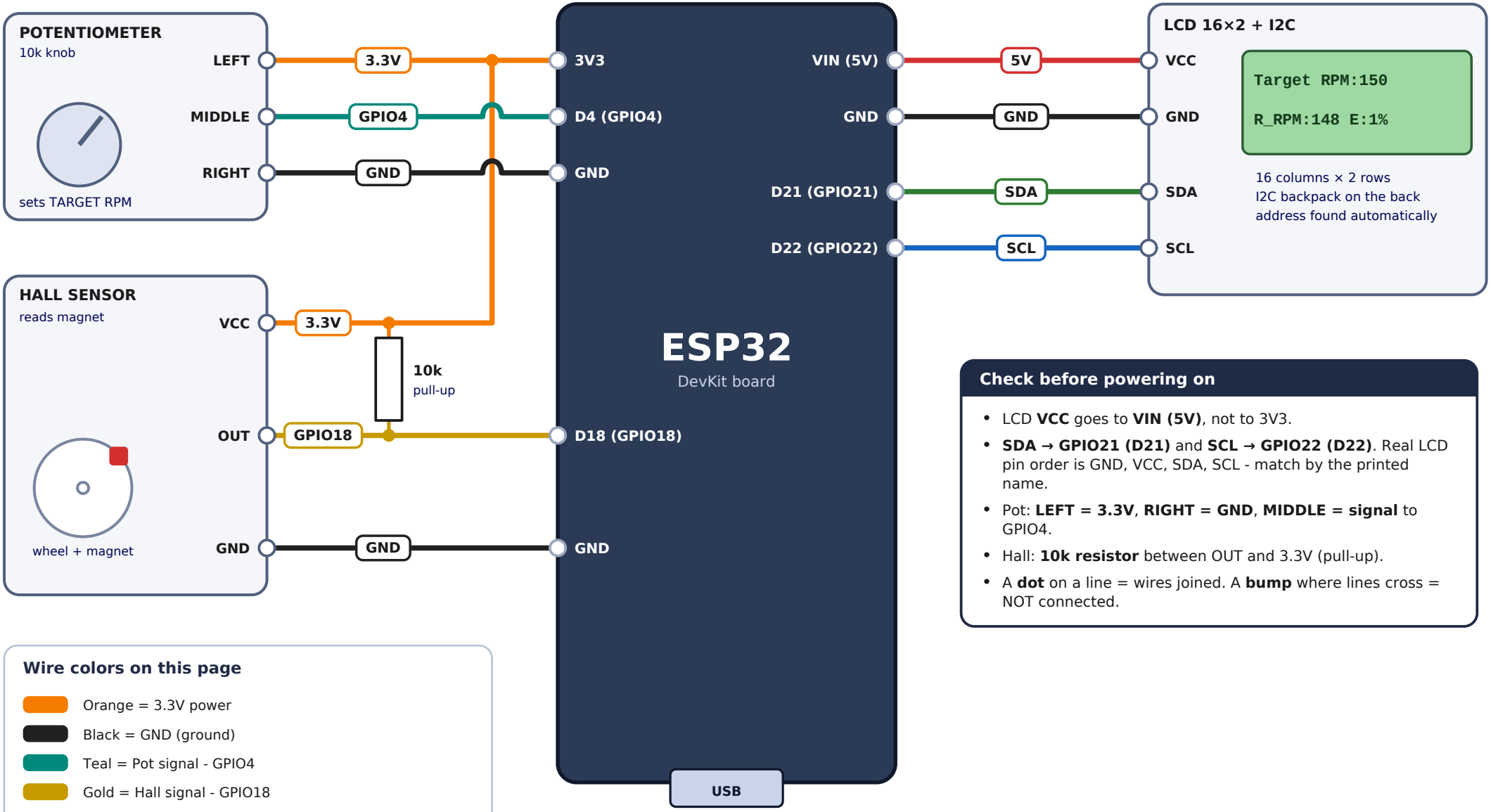
Pot sets TARGET RPM → Hall sensor measures REAL RPM → ESP32 changes motor PWM until they match → LCD shows Target / R_RPM / E %



Wire colors used in this guide (use the same colors on your real wires)

- **Orange** — 3.3V power
- **Red** — 5V power (VIN)
- **Purple** — +12V battery
- **Black** — GND (ground)
- **Green** — SDA (I2C data) - GPIO21
- **Blue** — SCL (I2C clock) - GPIO22
- **Teal** — Pot signal - GPIO4
- **Gold** — Hall signal - GPIO18
- **Brown** — IN1 - GPIO2
- **Pink** — IN2 - GPIO15
- **Cyan** — ENA (PWM speed) - GPIO5
- **Indigo** — Motor wire 1 (OUT1)
- **Lime** — Motor wire 2 (OUT2)

Part A - ESP32 ↔ Potentiometer, Hall sensor, LCD



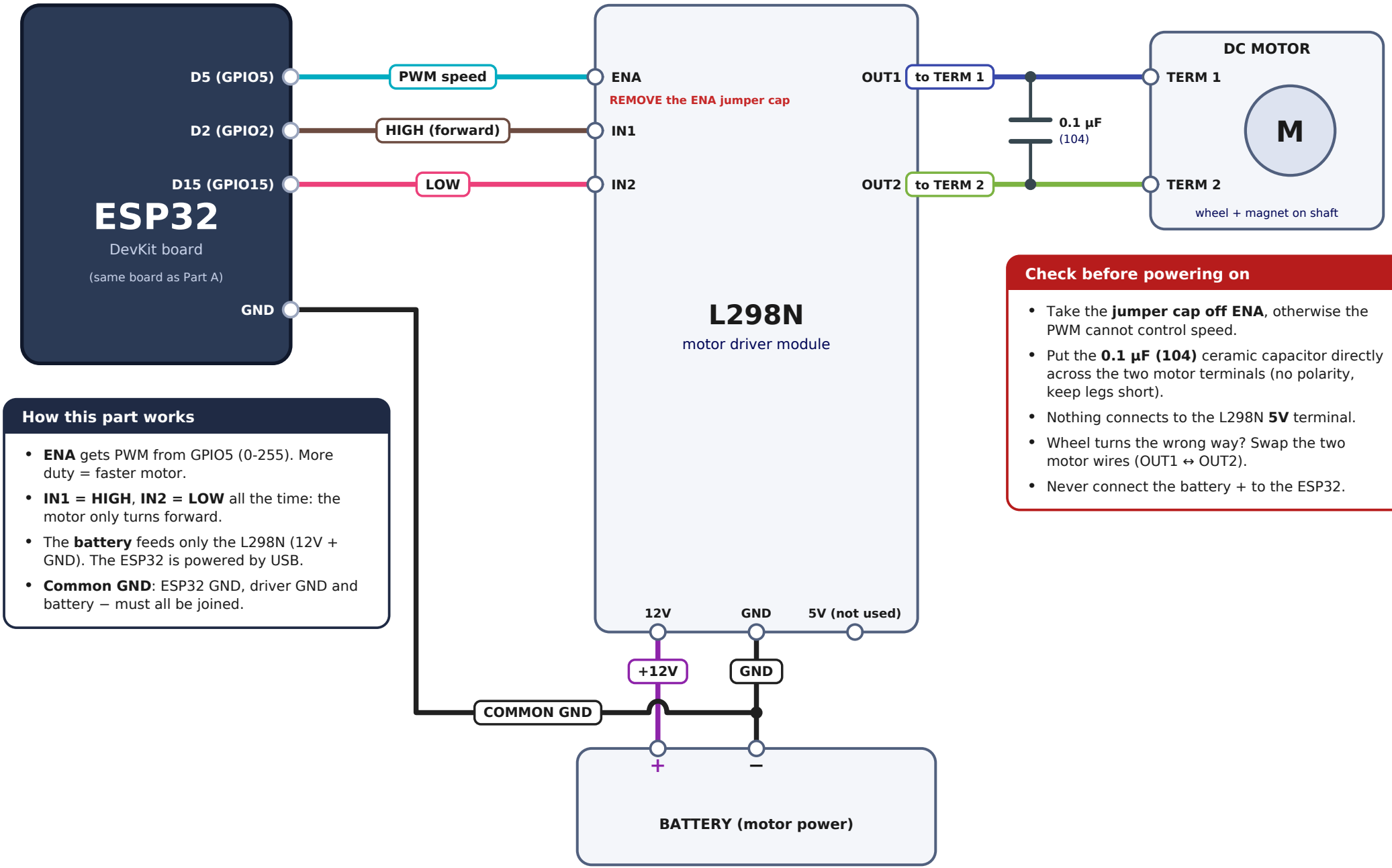
Check before powering on

- LCD **VCC** goes to **VIN (5V)**, not to 3V3.
- **SDA → GPIO21 (D21)** and **SCL → GPIO22 (D22)**. Real LCD pin order is GND, VCC, SDA, SCL - match by the printed name.
- Pot: **LEFT = 3.3V**, **RIGHT = GND**, **MIDDLE = signal** to GPIO4.
- Hall: **10k resistor** between OUT and 3.3V (pull-up).
- A **dot** on a line = wires joined. A **bump** where lines cross = NOT connected.

Wire colors on this page

- Orange = 3.3V power
- Black = GND (ground)
- Teal = Pot signal - GPIO4
- Gold = Hall signal - GPIO18
- Red = 5V power (VIN)
- Green = SDA (I2C data) - GPIO21
- Blue = SCL (I2C clock) - GPIO22

Part B - ESP32 ↔ L298N driver, Motor, Battery



How this part works

- **ENA** gets PWM from GPIO5 (0-255). More duty = faster motor.
- **IN1 = HIGH**, **IN2 = LOW** all the time: the motor only turns forward.
- The **battery** feeds only the L298N (12V + GND). The ESP32 is powered by USB.
- **Common GND**: ESP32 GND, driver GND and battery − must all be joined.

Check before powering on

- Take the **jumper cap off ENA**, otherwise the PWM cannot control speed.
- Put the **0.1 µF (104)** ceramic capacitor directly across the two motor terminals (no polarity, keep legs short).
- Nothing connects to the L298N **5V** terminal.
- Wheel turns the wrong way? Swap the two motor wires (OUT1 ↔ OUT2).
- Never connect the battery + to the ESP32.

Connection list - every wire, one by one

#	FROM (component → pin)	TO (component → pin)	Wire	Note	Done
POTENTIOMETER (knob = target RPM)					
1	**Pot LEFT pin**	**ESP32 3V3**	**Orange**	Power for the pot	☐
2	**Pot MIDDLE pin (wiper)**	**ESP32 D4 (GPIO4)**	**Teal**	Signal. D4 is an ADC2 pin - fine, WiFi is not used	☐
3	**Pot RIGHT pin**	**ESP32 GND**	**Black**	If the knob works backwards, set INVERT_POT = false in the sketch	☐
HALL SENSOR (reads the magnet on the wheel)					
4	**Hall VCC**	**ESP32 3V3**	**Orange**	3.3V, not 5V	☐
5	**Hall GND**	**ESP32 GND**	**Black**		☐
6	**Hall OUT**	**ESP32 D18 (GPIO18)**	**Gold**	Signal pulse once per wheel turn (1 magnet)	☐
7	**10k resistor - leg 1**	**Hall OUT (same point as the wire above)**	**Gold**	Pull-up resistor	☐
8	**10k resistor - leg 2**	**ESP32 3V3 (same point as Hall VCC)**	**Orange**	Pull-up resistor	☐
LCD 16×2 WITH I2C BACKPACK					
9	**LCD GND**	**ESP32 GND**	**Black**		☐
10	**LCD VCC**	**ESP32 VIN (5V)**	**Red**	5V, not 3.3V	☐
11	**LCD SDA**	**ESP32 D21 (GPIO21)**	**Green**	Changed: SDA is now GPIO21	☐
12	**LCD SCL**	**ESP32 D22 (GPIO22)**	**Blue**	SCL stays on GPIO22	☐
L298N MOTOR DRIVER					
13	**L298N IN1**	**ESP32 D2 (GPIO2)**	**Brown**	Direction (always HIGH)	☐
14	**L298N IN2**	**ESP32 D15 (GPIO15)**	**Pink**	Direction (always LOW)	☐
15	**L298N ENA**	**ESP32 D5 (GPIO5)**	**Cyan**	PWM = speed. Remove the ENA jumper cap first	☐
16	**Battery +**	**L298N 12V terminal**	**Purple**	Motor power	☐
17	**Battery −**	**L298N GND terminal**	**Black**		☐
18	**ESP32 GND**	**L298N GND terminal**	**Black**	Common ground - join it at the same GND terminal as the battery −	☐
MOTOR					
19	**L298N OUT1**	**Motor terminal 1**	**Indigo**	Swap OUT1/OUT2 wires if it spins the wrong way	☐
20	**L298N OUT2**	**Motor terminal 2**	**Lime**		☐
21	**Motor terminal 1**	**Motor terminal 2 (via 0.1 µF / 104 capacitor)**	**Black**	Capacitor legs on the motor terminals, no polarity	☐

ESP32 pin map and important notes

ESP32 pin map

GPIO	Label	Function	Connected to	Wire
3V3	3V3	3.3V power out	Pot LEFT, Hall VCC, 10k resistor	**Orange**
VIN	VIN	5V (from USB)	LCD VCC	**Red**
GND	GND	Ground (several pins)	Pot RIGHT, Hall GND, LCD GND, L298N GND	**Black**
GPIO4	D4	Analog input (ADC2)	Pot MIDDLE	**Teal**
GPIO18	D18	Digital input	Hall OUT (+ 10k pull-up to 3.3V)	**Gold**
GPIO21	D21	I2C SDA	LCD SDA	**Green**
GPIO22	D22	I2C SCL	LCD SCL	**Blue**
GPIO2	D2	Output	L298N IN1	**Brown**
GPIO15	D15	Output	L298N IN2	**Pink**
GPIO5	D5	PWM output	L298N ENA	**Cyan**

Matching lines in the sketch

- `#define SDA_PIN 21` and `#define SCL_PIN 22` - already correct in your code.
- Also change the comment at the top of the sketch: `LCD: SDA -> D21, SCL -> D22` (it still says D23).
- `POT_PIN 4, HALL_PIN 18, IN1 2, IN2 15, ENA 5`

Important notes

- **All grounds joined:** battery −, L298N GND, ESP32 GND. Pot, Hall and LCD GND go to the ESP32 GND.
- **Two voltages:** Pot and Hall use **3.3V**. LCD uses **5V (VIN)**. Motor uses the battery.
- **Remove the ENA jumper cap** on the L298N, or the speed cannot be controlled.
- **10k pull-up** from Hall OUT to 3.3V is needed for a clean pulse.
- **0.1 µF (104)** capacitor across the motor terminals reduces noise that disturbs the Hall signal.
- **ESP32 is powered by USB.** The battery goes only to the L298N. Never connect the battery + to the ESP32.
- **Upload fails?** GPIO2 and GPIO15 are boot pins. Unplug the IN1 wire while uploading, then plug it back.
- **LCD blank?** Check SDA = 21 and SCL = 22, turn the contrast screw on the backpack. The Serial Monitor prints "LCD not found" if the wiring is wrong.
- **Knob works backwards?** Set `INVERT_POT = false`. **Wheel spins the wrong way?** Swap the two motor wires.
- **Wrong RPM shown?** Use the Serial Monitor commands: `p<N>` (magnets), `t<sec>` or `r<RPM>` (115200 baud).